

ST CLAIR COUNTY INTL, MI, USA

WMO: 725384

Lat: 42.911N

Lon: 82.529W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.2	-22.6	0.5	-17.8	-19.9	0.7	-14.4	9.3	-2.6	8.4	-3.1	1.6	300	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	32.0	23.0	29.9	21.9	27.9	20.9	24.3	29.0	23.2	27.6	22.4	26.6	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	26.5	22.2	17.3	25.9	21.1	16.2	25.0	74.3	28.8	70.0	27.4	66.7	26.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.5	DB	-23.2	34.1	4.5	2.0	-26.5	35.5	-29.1	36.6	-31.7	37.7	-35.0	39.1	(4)
(5)			WB	-22.8	26.3	4.9	1.3	-26.3	27.2	-29.2	28.0	-31.9	28.7	-35.5	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-4.2	-3.6	1.1	7.3	13.6	18.5	20.9	20.0	16.5	10.3	4.2	-1.0	(6)	
	DBStd	10.11	6.21	6.13	5.92	4.89	4.68	3.89	3.28	3.05	4.07	4.68	4.72	5.02	(7)	
	HDD10.0	1816	441	382	284	109	16	1	0	0	2	54	184	343	(8)	
	HDD18.3	3777	699	615	535	331	161	45	12	18	83	253	425	601	(9)	
	CDD10.0	1338	0	1	7	29	128	256	338	309	196	64	9	1	(10)	
	CDD18.3	257	0	0	0	1	14	50	92	69	27	4	0	0	(11)	
	CDH23.3	2565	0	0	3	22	187	541	901	608	268	36	0	0	(12)	
	CDH26.7	728	0	0	0	3	45	170	281	149	72	7	0	0	(13)	
(14) Wind	WSAvg	2.7	3.2	3.1	3.2	3.3	2.7	2.2	1.9	1.8	2.1	2.6	3.0	3.0	(14)	
(15) Precipitation	PrecAvg	857	58	49	60	77	93	77	86	79	84	65	66	62	(15)	
	PrecMax	1194	99	99	129	158	194	151	171	156	178	170	139	112	(16)	
	PrecMin	627	18	13	23	21	41	22	14	16	15	16	21	21	(17)	
	PrecStd	123	23	24	28	32	38	36	40	31	47	32	29	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	14.0	22.5	26.3	30.2	32.6	33.0	32.5	32.2	27.3	19.0	14.1	(19)	
		MCWB	10.9	10.1	15.1	16.6	20.5	23.3	23.7	23.7	22.4	20.0	13.7	11.9	(20)	
	2%	DB	9.1	9.9	17.3	22.4	27.7	31.0	32.0	29.9	28.7	23.7	16.3	11.2	(21)	
		MCWB	7.9	7.6	11.7	14.2	19.3	22.2	23.5	22.5	21.3	17.6	12.6	9.6	(22)	
	5%	DB	6.3	7.0	13.6	19.0	25.2	28.7	29.7	27.9	26.6	21.2	13.9	7.8	(23)	
		MCWB	4.3	4.5	9.4	12.5	17.8	21.1	22.2	21.2	19.7	16.0	10.8	6.0	(24)	
	10%	DB	3.7	3.9	10.1	16.3	22.6	27.0	27.8	27.1	24.0	18.1	12.1	5.9	(25)	
		MCWB	2.3	1.9	6.9	10.8	16.1	20.0	21.1	20.6	18.4	14.5	9.4	4.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.9	15.9	18.7	22.6	24.6	26.4	25.3	24.4	21.0	15.4	12.8	(27)	
		MCDB	12.2	13.0	21.4	24.1	27.9	30.5	31.9	30.5	29.3	25.0	17.5	14.3	(28)	
	2%	WB	8.0	7.7	12.6	15.6	20.7	23.2	24.5	23.7	22.6	19.0	13.3	9.8	(29)	
		MCDB	8.6	9.2	15.8	20.2	25.7	28.5	29.1	27.7	26.6	22.1	15.8	11.0	(30)	
	5%	WB	4.8	4.3	10.0	13.5	19.2	22.2	23.4	22.7	21.2	16.9	11.4	6.5	(31)	
		MCDB	6.2	6.2	13.5	18.2	23.6	27.1	27.6	26.5	24.2	19.7	13.7	7.7	(32)	
	10%	WB	2.6	2.3	7.1	11.4	17.3	21.0	22.4	21.7	19.8	15.2	9.4	4.1	(33)	
		MCDB	3.8	4.0	9.6	15.7	21.3	25.1	26.6	25.5	23.0	18.1	11.5	5.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	9.5	11.2	12.3	12.4	12.3	11.8	12.5	10.7	8.4	6.8	(35)	
		MCDBR	9.1	11.0	14.7	16.3	15.5	14.9	13.8	13.3	14.9	14.5	11.4	9.3	(36)	
	5% WB	MCWBR	8.2	9.0	9.9	9.7	8.5	7.4	6.7	6.8	8.0	8.5	8.7	8.0	(37)	
		MCWBR	8.7	10.3	13.6	14.3	13.5	12.8	11.8	11.3	12.3	11.8	10.3	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.323	0.340	0.386	0.410	0.434	0.422	0.411	0.379	0.358	0.319	0.295	(40)	
		taud	2.345	2.295	2.369	2.267	2.253	2.219	2.277	2.295	2.372	2.408	2.485	2.444	(41)	
	Ebn at Noon		821	869	901	879	864	842	848	845	848	822	811	810	(42)	
		Edh at Noon	82	102	107	129	134	139	130	124	107	91	72	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.32	2.18	3.46	4.57	5.37	6.10	6.01	5.23	4.19	2.68	1.61	1.11	(44)	
	RadStd		0.17	0.21	0.34	0.40	0.46	0.35	0.31	0.23	0.37	0.23	0.20	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page